

INSPACE-CAM-2026 Collision Avoidance Filing

Filing Reference ID:	6
Primary Spacecraft:	ISS (ZARYA) (NORAD 25544)
Secondary Spacecraft:	NORAD 900
Operator of Record:	OrbitGuard Swarm Agent Orchestrator
Time of Closest Approach (TCA):	2026-08-19 11:02:54.787509
Filing Status:	Filed
Submission Timestamp:	2026-08-20 01:39:29.118382

Regulatory Briefing & Maneuver Justification

****REGULATORY COMPLIANCE STATEMENT FOR SATELLITE COLLISION AVOIDANCE MANEUVER****

****Introduction****

This regulatory compliance statement is issued in accordance with applicable regulations and guidelines to inform stakeholders of the necessary measures taken to mitigate a potential collision between the International Space Station (ISS) ZARYA (NORAD 25544) and CALSPHERE 1 (NORAD 900). The purpose of this statement is to provide transparency and compliance with relevant regulations, ensuring the safety of both satellites and the space environment.

****Hazard Mitigation****

A potential collision between the ISS ZARYA and CALSPHERE 1 is imminent, with a predicted Time of Closest Approach (TCA) of 2026-08-19T11:02:54.787509. The calculated Miss Distance is 0.35 km, and the Collision Probability (Pc) is 0.000145. To mitigate this hazard, a satellite collision avoidance maneuver has been executed to ensure the safe separation of the two satellites.

****Target Safety Window****

The target safety window for this maneuver is as follows:

- * TCA: 2026-08-19T11:02:54.787509
- * Miss Distance: 0.35 km
- * Collision Probability (Pc): 0.000145

The safety window is critical to ensuring the avoidance of a potential collision. Any deviation from this window may result in a hazardous situation.

Astrometric State Vectors & Covariance Matrices (TCA Epoch)

Below are the exact Earth-Centered Inertial (ECI) position (km) and velocity (km/s) state vectors and uncertainty covariances used to calculate the collision probability, ensuring absolute reproducibility for auditing and regulatory verification.

Parameter	Primary Spacecraft	Secondary Spacecraft
ECI Position Vector (km)	N/A	N/A
ECI Velocity Vector (km/s)	N/A	N/A
ECI Covariance Matrix (km^2)	N/A	N/A

Authorization & Sign-off

Authorized Operator Signature

Date